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Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
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Zip Code:	29425-2203	**Congressional District:**	1
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Task Description:	Extensive pathology and mortality data from Argonne National Laboratory for B6CF1 mice receiving single and protracted exposure to neutrons will be used to estimate radiation-induced risk (mortality and morbidity) for heavy ions. The justifications for using neutron data as a surrogate for HZE particles are that the 0.85 MeV JANUS neutrons are a high LET radiation, and their biological effectiveness and dose-response at lower doses have been shown to be similar to those for Fe and Argon. Although we will acquire heavy ion mortality data as it becomes available, the neutron data will be used to fit both empirical and biologically motivated dose-response models. Empirical models will be examined because they have already been applied to these data in order to successfully make interspecies predictions (mouse to dog to human) of radiation-induced mortality for low LET radiation (both acute and protracted). The biological models will be examined to not only compare their predictions to the empirical models, but also to examine how knowledge of the pathogenesis of disease processes affects quantitative models used to predict radiation-induced mortality risks.		

	Finally, the results of this modeling project will be synthesized into practical estimates of the potential radiation risks for astronauts on prolonged space flights. To obtain neutron RBE values for a variety of cancer and cardiovascular endpoints. Evaluate the assumption of a constant RBE over dose levels. Examine the predictability of gamma exposed animals to human cancer risk. Will provide needed information for estimating astronaut cancer risks from HZE exposures in space.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Help to set exposure limits for workers exposed to high LET radiation.
Task Progress:	Cardiovascular Effects: Of concern to NASA is the possible non-cancer effects of radiation exposure such as heart disease. After a long latency period increased mortality from heart disease has been observed in the A-bomb survivors. Very detailed cardiovascular pathology was carried out for one of the major Janus experiments involving both gamma and neutron exposures. Our analysis suggests that there is a neutron effect greater than what is observed with gamma exposures. Also it appears that low doses of neutron can affect CVD mortality while it is not clear that low-dose exposures of gamma in either the mouse or in humans is a CVD risk. Although CVD events are rarely lethal, our findings suggest that they are capable of contributing to radiation-induced mortality. The individual JANUS mouse studies involving a single exposure provide the strongest evidence for a radiation-induced cardiovascular response for both gamma rays and neutrons. Relatively low doses were administered in only two of the experiments which were neutron exposures below 0.1 Gy. There were, however, no consistent increases in CVD effects among the mice in a 60 once-weekly study. In the experiment which used single exposures administered only to female mice there were consistent increases in CVD effects at neutron doses < 0.10. We recognize that CVD in the human studies are examined from the perspective of mortality rather than incidence. However, our analyses suggest that while incidence is a weaker response, it does tend to trend with mortality. Despite being variable and uncertain at low doses, the radiation-induced CVD risks reported in the human studies and our mouse study are consistent. The B6CF1 mouse appears to have an excess relative risk for gamma-ray exposure that is similar to that observed in the A-bomb survivors and nuclear workers. In all of these studies the excess risk is weak in comparison to neoplastic endpoints. As such, not much can be said about potential CVD effects at gamma-ray doses less than 0.5 Sv that are of concern for environmental and occupational exposures. However, if the RBE for CVD effects is about 5 to 8 in the mouse as we observed in the female mouse irradiated with fission neutrons showed limited increases in CVD risk at several doses that are well below 0.10 Gy (i.e. 0.5 Sv). In summary, radiation induces CVD mortality in mice in a dose-dependent manner. This induced mortality arises from either gamma ray or neutron exposures. Further, increased CVD was observed at low neutron exposures (<0.1 Gy) among the female mice exposed to single doses. Although the RBE for neutrons appears to be around 5 to 8 for single exposures, the uncertainties in these data make this a tenuous estimate. Nevertheless, the neutron effects are generally greater than those observed from gamma exposures. In addition, females appear to be more susceptible than males to radiation-induced CVD effects. Within the limits of these data, the risk of CVD may be enhanced when doses are protracted, but the data are too variable to warrant estimates of this effect. If the CVD effects observed in these experiments are biologically related to human CVD, then the radiation-induced CVD risk for humans receiving protracted exposure to high LET radiation (e.g., astronauts) may be significantly higher than is currently being estimated from the A-bomb survivors who received a single exposure primarily to gamma radiation. Cancer Effects – Empirical Methods: Cox regression has been applied to the specific subcategories of the lymphoreticular cancers. The major category is the lymphosarcomas followed by the reticulum cell sarcomas with their two subcategories. What is observed in the analyses is that for acute single exposures there is no apparent effect typically before 1 Gy of gamma exposure while there is a linear increase with no apparent threshold for neutrons. The two leukemia types myeloid and lymphocytic have relatives few cases and there is not an apparent relationship with exposure. This detailed analysis of the JANUS mouse data for lymphoreticular cancers will be completed soon. The analysis of solid tumors will then follow. The focus will be on estimation of risk and RBEs and DREFs estimates if possible. If there are no low dose effects for gamma then an RBE has limited meaning. Cancer Effects – Biologically Based Models In recent years, biologically-based models have been used to estimate cancer effects in both animal and human studies. This has included the A-bomb survivors with single acute exposures as well as uranium miners with continuous radon exposures. Use of the biologically-based two stage clonal expansion (TSCE) model and its possible extensions promises to provide a better tool for cancer risk estimation. The model is essentially a stochastic description of the mutation of normal cells into intermediate cells which undergo division and death. This is followed by a second mutation to malignancy and subsequent cell division until tumor detection. A number of investigators have applied this model to the A-bomb data, as well as radon exposure among miners, and animal studies of radon and radium. Applications have, over time, increased in complexity and are reliable models for relating radiation exposure to human cancer risk. If radiation affects only the initial mutation rate then the incidence rate describes the usual proportional hazard model that is used in the analysis of the epidemiology data. We used the broad cancer groupings of lymphoreticular and solid tumors and fit the TSCE model to the JANUS gamma exposed mouse data using only doses less than 4 Gy. Programs for this analysis were developed using MATLAB software. The first observation is that there is good agreement in the parameter estimates between the male and female mice. Next, the dose effects the second stage mutation rate in both tumor groups and also effects the first mutation rate in the solid tumors but not the lymphoreticular tumors. The relative risks of the two tumor groups for the female mice as a function of age show that the solid tumors are fairly constant over time compared with the lymphoreticular tumors. This is similar to what is observed in the A-bomb studies. The biologically based TSCE model needs changes before comparisons are appropriate between gamma and neutron as well as humans. The current model described above assumes that the mutation rates are permanently changed by the exposure. This certainly cannot be the case for mutations other than the initial one affecting normal stem cells. Once this model extension is completed the mouse data will be analyzed and compared with the human gamma data for purposes of estimating neutron effects in human under different exposure scenarios.

Bibliography Type:	Description: (Last Updated: 02/25/2014)
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Papers from Meeting Proceedings	Hoel DG, Carnes BA. "Cancer Dose-Response Analysis of the Radium Dial Workers." HEIR 2004 9th International Conference on Health Effects of Incorporated Radionuclides - Emphasis on Radium, Thorium, Uranium and their Daughter Products, GSF-National Research Center, Neuherberg, Germany, November 29-December 1, 2004. Proceedings from the HEIR 2004 9th International Conference on Health Effects of Incorporated Radionuclides - Emphasis on Radium, Thorium, Uranium and their Daughter Products, November 29-December 1, 2004. GSF-Bericht : Institut für Strahlenschutz, 2005, p. 169-173. , Jun-2005